



Predicting Adolescent Suicidal Risk from Multi-task-based Speech: An Ensemble Learning Approach

Xi Chen¹, Renzhe Yu¹, Yansen Tan¹, Yiyi Li¹, Quan Qian², Ying Lin¹

¹Department of Psychology, Sun Yat-sen University, China

²Guangzhou Zhengxin Psychological Counseling Co. Ltd., China
linyings23@mail.sysu.edu.cn

Abstract

Adolescent suicide is a pressing global public health issue. Timely identification of suicide risk is crucial. Traditional methods of assessing suicide risk are often limited by their reliance on subjective input and resource requirements. This paper aims to address these limitations by detecting suicide risk from multi-task-based speech, utilizing a dataset of 600 Chinese adolescents (age: 10-18 yr) provided by the 1st SpeechWellness Challenge. Our approach involved both acoustic and semantic features extracted by OpenSmile, Emotion2Vec, and a fine-tuned BERT-Chinese model. The base models were trained using XGBoost and SVM, etc., with hyperparameters tuned by Bayesian optimization. Then we implemented a multi-model and multi-task nested voting ensemble framework to integrate the base models, achieving a final test set accuracy of 0.63 (recall=0.74, F1≈0.67). This work highlights the potential of voice-based biomarkers in mental health assessment.

Index Terms: adolescent suicide, speech-based predictive modeling, ensemble learning

1. Introduction

Research on adolescent suicide prediction is of importance due to its potential to inform prevention strategies and mitigate the risk of premature mortality among this vulnerable population. Identifying reliable predictors of suicide can facilitate early intervention, reducing the likelihood of suicide completion.

The existing methods for assessing suicidal risk, including self-report, clinical assessment, are effective in some cases. However, they also have a number of limitations. First, self-report methods are susceptible to factors such as adolescents' emotional fluctuations and social expectations, resulting in more subjective data. Second, traditional clinical assessments rely on diagnosis by professionals and are limited by time, resources, and regional availability. In addition, the factors influencing suicidal ideation are complex and multidimensional, involving individual and environmental aspects. It is often difficult for traditional methods to comprehensively capture such diverse features, leading to inaccurate or one-sided assessment results. It is thus difficult for traditional single assessment methods to fully capture risk factors [1].

To address the above limitations, speech-based prediction becomes a promising solution. The 1st SpeechWellness Challenge (SW1) in InterSpeech 2025 offers a speech dataset of 600 participants and reveal each participant's responses to three different speech tasks designed to capture a range of linguistic, cognitive, and emotional markers relevant to suicidal risk detection [1]. In particular, Task 0 (ER) assesses

spontaneous speech, providing insights into emotional expression, coping strategies, and linguistic patterns associated with distress. Task 1 (PR) offers a controlled linguistic task, enabling the analysis of prosody, articulation, and speech rhythm. Task 2 (ED) evaluates emotional perception and cognitive processing by requiring participants to describe a facial expression. Deficits in emotion recognition and verbalization are commonly linked to suicidal risk, making this task valuable for assessing affective and social cognition. As such, the SW1 dataset facilitates a comprehensive analysis of semantic, prosodic, and cognitive-linguistic features, enhancing the reliability of speech-based suicidal risk assessment.

By combining different models derived by different methods, features and data, ensemble learning can compensate for the shortcomings of a single model and integrate the advantages of different models to reduce overfitting and improve generalizability. Considering the multi-task nature of the SW1 dataset and the high complexity of speech data, ensemble learning offers a straightforward and promising way to build a competitive suicide risk prediction model. In particular, ensemble learning allows us to involve richer input information by combining models based on acoustic features (e.g., emotional tone, speech rate, etc.) and semantic features (e.g., textual utterance analysis). It also enhances the model's ability to capture complex affective and psychological states through joint learning of multiple tasks.

2. Methods

In this section, we elaborated on the proposed method for predicting the adolescent suicide risk based on the multi-task speech data offered by SW1. For details of the dataset and the third-party tools used in the method, refer to Section 1.1, 1.2 and 1.3 in Supplementary Materials.

2.1. Feature engineering

2.1.1. Independent speech features

In this study, we conducted a statistical analysis of audio features across three different tasks (TASK0, TASK1 and TASK2) to identify significant features in suicidal risk detection. We utilized the openSMILE [2, 3] toolkit to extract a total of 88 eGeMAPS[1] features from the training set of each task. To assess the differences in features between the two categories (Label as 1 and 0), we performed independent two-sample t-tests on each feature. In each task, features with a p-value less than 0.05 were considered statistically significant. The results indicated that there were 10, 12, and 11 significant features for TASK 0, 1, and 2, respectively.

In addition, we extracted frequency band energy features from the audio signals to capture audio characteristics related to suicidal risk. First, we loaded the audio data using the librosa [5] and represented the frequency characteristics of the audio signal using a Mel Spectrogram. The Mel Spectrogram is obtained by initially applying the Short-Time Fourier Transform (STFT) to the audio signal as follows: :

$$S[m, k] = \left| \sum_{n=0}^{N-1} y[n]w[n - mH]e^{-\frac{j2\pi kn}{N}} \right|^2 \quad (1)$$

where $S[m, k]$ is the spectrum of the m -th time frame and the k -th linear frequency, $y[n]$ is the audio signal, $w[n]$ is the window function, H is the hop length, and N is the window length.

Then converting the frequency axis to the Mel scale and taking the logarithm of the Mel spectrogram, we obtained the Mel logarithmic spectrum $M_{db}[m, i]$, where i is the index of the frequency band. We then averaged the energy of each frequency band in the $M_{db}[m, i]$ along the time dimension to generate frequency band energy features:

$$freq[i] = \frac{1}{T} \sum_{m=1}^T M_{db}[m, i] \quad (2)$$

where $freq[i]$ is the average energy of the i -th frequency band in $M_{db}[m, i]$, and T is the amount of time frames.

To analyze features across different frequency ranges, we divided the extracted frequency band energy into low frequency (0-40 Hz), mid frequency (40-80 Hz), and high frequency (80-128 Hz) bands, and calculated statistical features for each band, including mean, standard deviation, maximum, and minimum. Additionally, we calculated the overall energy's mean, standard deviation, range, skewness, and kurtosis to comprehensively describe the distribution of frequency band energy features. Taken together, we extracted 145 features, including 128 individual frequency band energies, 12 statistical features from the three frequency bands, and 5 overall statistical measures.

Based on the training set, we selected the features with significant inter-group difference ($p < 0.05$, effect size > 0.15). Then we performed cross-validation using XGBoost to further select key features. Ultimately, the three tasks identified 5, 4, and 8 key features for TASK 0, 1, and 2, respectively, which are of great significance for building suicide recognition models.

Furthermore, we enriched the acoustic feature set by incorporating a broader range of features extracted using openSMILE, including spectral, cepstral, and voice quality parameters, along with emotion embeddings derived from Emotion2Vec and a novel silence ratio feature. This process resulted in a high-dimensional feature space with 725,931 features, which were then standardized and subjected to an independent-sample t-test to identify significant features ($p < 0.05$). Principal Component Analysis (PCA) was employed to reduce the dimensionality while preserving 99% of the variance. This approach ensured that the most salient acoustic features were retained for subsequent analyses.

2.1.2. Integrated speech features

We proposed a framework for feature fusion and optimization. We merged the aforementioned independent speech features according to the tasks. The initial feature dimensions were 15, 16, and 19, respectively. To extract more effective feature representations, we applied Incremental Principal Component Analysis (Incremental PCA) for dimension reduction of the features for each task. Considering the inter-task correlations,

we utilized a feature interaction method based on Partial Least Squares Regression (PLSR) to handle task pairs (TASK0-TASK1, TASK0-TASK2, and TASK1-TASK2) separately.

Specifically, for each pair of tasks, we first used PLSR to learn the mapping relationship between the feature spaces of the two tasks, extracting 4-dimensional interaction components. We then concatenated these interaction components with the first 4 original features of the target task to form 8-dimensional mixed interaction features. This design retains the direct information of the original features while capturing the latent associative patterns between tasks.

Finally, we combined all the base features and interaction features to obtain a 48-dimensional vector and performed recursive feature elimination with cross-validation (RFECV) based on Random Forest. Ultimately, we selected 40 of the most discriminative features, significantly reducing the dimensionality of the feature space while preserving the ability of the features to discriminate. In addition, by including gender and age, we had a total of 42 features.

2.1.3. Semantic features

We conducted feature engineering for the text classification task to optimize model performance and enhance generalization ability. Since the original data was provided in audio format, we first used the Paraformer speech recognition model to convert the audio into text, forming the input features for subsequent natural language processing (NLP) tasks. This conversion process not only preserved the semantic information but also enabled the BERT pre-trained model [6] to effectively extract features from the text data.

In the text processing stage, we utilized the BERT-Base-Chinese pre-trained model to encode the text. BERT constructs contextual relationships through a bidirectional Transformer mechanism and generates high-dimensional semantic features. Specifically, we used the hidden state vector at the [CLS] position as the sentence-level representation, which has a dimensionality of 768 and represents the semantic information of the entire sentence.

To further enhance the model robustness, we introduced the nlpcda data augmentation strategy. This strategy includes methods such as synonym replacement, character swapping, and random deletion to expand the training data and improve the model's generalization ability. Through this augmentation strategy, we can generate more variants of the training samples, increasing the model's adaptability to different inputs.

2.2. Model training and ensemble

2.2.1. XGBoost Training with Independent Features

We trained an XGBoost classifier in a 5-fold cross-validation (CV) on each set of speech features (please refer to section 2.1.1). The Bayesian optimization method was employed for tuning the model's hyperparameters (500 iterations with 50 random initialization), with the optimization objective of maximizing the CV accuracy.

2.2.2. Ensemble training with integrated features

As for integrated speech features (in section 2.1.2), we proposed an ensemble learning-based classification framework combining random forests and support vector machines (with both linear and RBF kernels). In the preprocessing stage, we standardized features and enhanced model robustness through

data augmentation, specifically by adding Gaussian noise ($\sigma = 0.01$) and applying scale perturbation (0.95-1.05).

The model training followed a two-phase strategy: first optimized the base classifiers' hyperparameters by Bayesian optimization (100 iterations with 10 random initializations), then determined the ensemble weights (50 iterations with 10 random initializations) to maximize 5-fold CV accuracy.

2.2.3. MLP training with semantic features

The semantic features (in section 2.1.3) were fed into a multi-layer perceptron (MLP) classifier, which comprises two fully connected layers with ReLU activation functions. To mitigate the overfitting risk, dropout regularization was employed. The output layer utilized a sigmoid activation function to perform binary classification.

For the loss function, we employed Focal Loss to address class imbalance issues. Additionally, AdamW was used as the optimizer, with a learning rate set to $2e-5$. A cosine annealing learning rate scheduler was incorporated to enable dynamic adjustment of the learning rate throughout the training process.

During training, we utilized batch-wise optimization strategy with the batch size of 16. To enhance the training stability, mixed precision training was applied, facilitating a faster training process while reducing memory consumption. We split the dataset into 80% for training and 20% for validation, performing evaluations on the validation set at the end of each epoch to assess the model's generalization ability.

2.2.4. Model ensemble

The above three types of models were ensembled through a hierarchical approach. First, we conducted hard voting on the XGBoost models for each set of independent speech features, and then combined their results with those from the semantic feature model and the integrated feature model using additional hard voting. All models were trained and validated using 5-fold cross-validation on the training set. The final performance was evaluated on the given test set of 100 samples using accuracy, recall, precision, and F1-score as evaluation metrics.

2.3. Model Explanation

To understand the contributions of different acoustic features and their interactions, we conducted a comprehensive feature importance analysis based on our ensemble model. The importance scores were calculated from three perspectives and then combined using the optimized weights obtained from ensemble training. Specifically, for the random forest classifier, we computed the mean decrease in Gini impurity across all trees. For the linear SVM, we used the absolute values of the feature coefficients. For the RBF SVM, we employed permutation importance, which measures the decrease in model performance when a feature is randomly shuffled. The final importance score for each feature was obtained by summing up these individual importance scores using their corresponding model weights determined in the ensemble optimization phase.

3. Results

3.1. Prediction performance

3.1.1. Performance of individual models

This section presents the classification performance of the XGBoost model, SVM and RF ensemble, and MLP model. As

shown in Table 1, the XGBoost model trained with Task 2 openSMILE features achieved an accuracy of 0.63. By incorporating multiple feature sets, XGBoost further improved its performance through a hard voting strategy, achieving an accuracy of 0.67 and an F1-score of 0.71, which was the best among all models in comparison. The SVM and RF ensemble approach, which capitalized on integrated speech features, also delivered stable results. The BERT+MLP model, leveraging semantic representations, achieved promising performance. These results highlight the usefulness of different feature sets and modeling strategies, with the XGBoost voting approach being the most effective in enhancing classification accuracy.

Table 1: Prediction performance of individual models*

Model / Features	Acc	Rec	F1
XGBoost / Task 0-openSMILE	0.61	0.58	0.6
XGBoost / Task 0-frequency band	0.62	0.56	0.6
XGBoost / Task 1-openSMILE	0.59	0.58	0.58
XGBoost / Task 1-frequency band	0.62	0.62	0.62
XGBoost / Task 2-openSMILE	0.63	0.62	0.63
XGBoost / Task 2-frequency band	0.58	0.58	0.58
XGBoost voting model / All above	0.67	0.78	0.71
SVM+RF / Integrated speech features	0.58	0.62	0.6
BERT+MLP / Semantic features	0.60	0.48	0.55

* Acc: accuracy; Rec: recall. The first six models listed in the table are trained on independent features in section 2.1.1 and are used as components of XGBoost voting model. The last two models are trained on features from section 2.1.2 and section 2.1.3 respectively.

3.1.2. Performance of ensemble models

We evaluated our ensemble model that ensembled ‘‘XGBoost voting’’, ‘‘SVM+RF’’, and ‘‘BERT+MLP’’ on both the validation and test sets. As shown in Table 2, the model achieved an accuracy of 0.63 on both datasets, with balanced performance on the validation set and higher recall on the test set, showing its effectiveness in capturing relevant patterns.

Table 2: Prediction performance of the final models*

Dataset	Acc	Pre	Rec	F1-score
Validation	0.63	0.63	0.62	0.63
Test	0.63	0.61	0.74	0.67

* Acc: accuracy; Pre: precision; Rec: recall.

3.2. Feature importance

In this study, the feature importance analysis was performed exclusively for TASK 2 (comprising features from openSMILE, emotion2vec, and silence ratio) for the following reasons. First, Task 2 offers a comprehensive set of features from multiple dimensions of speech, including time-domain, frequency-domain, and Mel-Frequency Cepstral Coefficients (MFCCs), which provide a rich array of data for evaluating the relationship between speech characteristics and suicidal risk. Unlike the other tasks that are more focused on specific aspects of speech (e.g., emotional regulation or facial expression recognition), Task 2 encompasses a broader spectrum of features that may be more indicative of an individual's emotional and cognitive states, thus offering greater potential for detecting subtle patterns of distress. Second, Task 2 has the potential to provide more nuanced insights into speech as a risk indicator. By

incorporating features from both emotion2vec (a speech emotion representation model) and traditional acoustic features, Task 2 captures not only raw speech characteristics but also higher-level emotional content. Given the complexity of suicidal tendencies, it is crucial to explore speech patterns from a multifaceted perspective. As Task 2 was the most comprehensive and potentially the most relevant to suicidal risk, it made sense to prioritize the analysis of features from this task.

In Task 2, the extracted features included time-domain features, frequency-domain features, and MFCCs, each contributing uniquely to the model's performance. For example, time-domain features like RMS energy capture variations in speech intensity, reflecting the emotional energy of the speaker. These features are important because fluctuations in speech intensity often correspond to changes in emotional states, such as distress. Similarly, frequency-domain features, such as Spectral Kurtosis and Spectral Slope, provide insight into the speech signal's frequency distribution. Spectral Kurtosis measures the sharpness of spectral peaks, which can indicate emotional intensity or distress, while Spectral Slope reveals the distribution of frequency components across the speech signal. Together, these features help the model understand both the energy and clarity of the speaker's emotional expression. Table 3 presents these ten most important features along with their corresponding importance values.

Table 3: Top 10 most important features

Feature	IMP
pcm_RMSEnergy_sma_risetime	0.002444
audSpec_Rfilt_sma_de[3]_quartile2	0.002306
pcm_fftMag_spectralKurtosis_sma_risetime	0.002069
mfcc_sma[12]_quartile2	0.002067
pcm_fftMag_fband250-650_sma_de_risetime	0.002052
pcm_fftMag_spectralSlope_sma_minRangeRel	0.002015
mfcc_sma[1]_iqr1-3	0.001893
pcm_fftMag_spectralCentroid_sma_de_peakRang	0.001879
eAbs	
logHNR_sma_linregerrQ	0.001843
mfcc_sma[5]_iqr2-3	0.001823

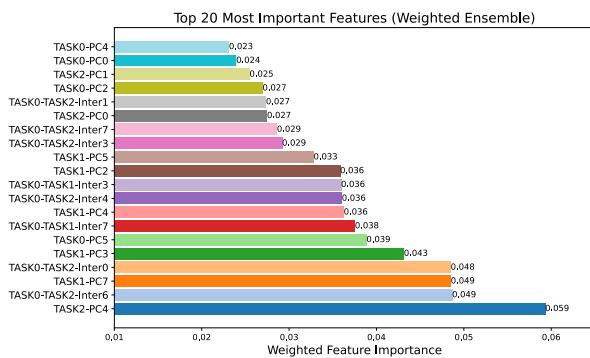


Figure 1: Feature importance in the ensemble model

Then, we conducted a further analysis of all features across the entire model. Figure 1 presents the feature importance analysis results from the weighted ensemble model. The results indicate that Task 2's PC4 contributes the most (importance score 0.059), highlighting its key role in suicide risk classification. Additionally, Task 2's interaction feature Inter6 (0.049) and Task 1's PC7 (0.049) also show high importance,

suggesting that speech features and cross-task interaction features are crucial for risk prediction. Overall, Task 2's PC features and Task 0-Task 2 interaction features are the most discriminative, emphasizing the critical role of multidimensional speech features in suicide risk detection.

4. Discussion and conclusion

4.1. Model performance

Our ensemble learning approach for predicting adolescent suicidal risk from multi-task speech data demonstrates promising performance in detecting psychological distress through speech patterns. BERT's semantic feature extraction effectively identifies language patterns associated with suicidal tendencies, including negative self-references, cognitive rigidity, and social isolation - traits previously documented in at-risk individuals [7, 8].

The multi-task approach, capturing different emotional, cognitive, and prosodic features, provides a comprehensive assessment of adolescents' mental states. Our ensemble method integrating diverse feature models outperforms individual classifiers, particularly in recall - crucial for identifying high-risk individuals. The model achieved 67% accuracy and 71% F1-score, indicating potential for practical applications, though further optimization is needed to improve detection of subtle risk indicators.

4.2. Feature importance

Feature importance analysis reveals that combinations of time-domain, frequency-domain, and MFCC features are critical for identifying suicidal risk. Time-domain features like RMS energy reflect speech intensity disruptions common in psychological distress. Spectral features, particularly spectral kurtosis and slope, indicate that speech clarity and emotional intensity serve as key risk indicators, with spectral kurtosis specifically linked to emotional distress markers.

MFCC features effectively capture speech timbre and prosody, which reflect emotional states like sadness and anxiety - common in at-risk adolescents. The interaction features between different speech tasks, especially between emotion regulation (Task 0) and facial expression description (Task 2), demonstrate strong discriminative ability. This suggests that relationships between emotional expression and social cognition provide crucial supplementary information for risk prediction.

4.3. Conclusion and future directions

This study demonstrates speech-based analysis's potential for predicting adolescent suicidal risk. By capturing subtle linguistic, prosodic, and emotional signals through multi-task learning and ensemble methods, our model offers a promising alternative to traditional assessment approaches. Early detection of distress markers through speech could become a valuable tool for timely intervention.

Future research may incorporate longitudinal data to track how speech characteristics evolve over time relative to stressors and mental health changes. This could provide deeper insights into individual vulnerability-environment interactions, potentially enabling more personalized intervention strategies. Using deep learning techniques may further optimize risk prediction, enhancing accuracy and real-world applicability.

5. References

- [1] Wen Wu, Ziyun Cui, Chang Lei, Yinan Duan, Diyang Qu, Ji Wu, Bowen Zhou, Runsen Chen, and Chao Zhang. “The 1st SpeechWellness Challenge: Detecting Suicide Risk Among Adolescents.” *arXiv preprint arXiv:2501.06474*, 2025.
- [2] F. Eyben, M. Wöllmer, and B. Schuller, “Opensmile: the munich versatile and fast open-source audio feature extractor,” in *Proceedings of the 18th ACM international conference on Multimedia*, 2010, pp. 1459–1462.
- [3] B. Schuller, S. Steidl, A. Batliner, et al., “The interspeech 2016 computational paralinguistics challenge: Deception, sincerity & native language,” in *Interspeech 2016*, 2016, pp. 2001–2005.
- [4] Eyben, F., Scherer, K.R., Schuller, B., Sundberg, J., André, E., Busso, C., Devillers, L., Epps, J., Laukka, P., Narayanan, S.S., & Truong, K.P., “The Geneva Minimalistic Acoustic Parameter Set (GeMAPS) for Voice Research and Affective Computing,” *IEEE Transactions on Affective Computing*, vol. 7, no. 2, pp. 190–202, Apr. 2016.
- [5] B. McFee, C. Raffel, D. Liang, M. McVicar, E. Battenberg, and O. Nieto, “librosa: Audio and music signal analysis in python,” in *Proceedings of The 14th Python in Science Conference*, 2015.
- [6] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” In *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.
- [7] S. Rude, E.-M. Gortner, and J. Pennebaker, “Language use of depressed and depression-vulnerable college students,” *Cognition & Emotion*, vol. 18, no. 8, pp. 1121–1133, Dec. 2004.
- [8] T. J. Ellis and B. E. Rutherford, “Cognition and Suicide: Two Decades of Progress,” *International Journal of Cognitive Therapy*, vol. 1, no. 1, pp. 47–68, Feb. 2008.